

Question			Marking Details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)		Any two (x1) from: • Low body temperature (1) • Metabolic rate low / example of (metabolic) reactions not required/ working at lower rate(1) • Less energy required for support / fish are buoyant (1) ORA for dog Reject dogs move more than fish/ dogs are more active than fish/ more oxygen needed for respiration		2		2		
	(e)		1. Mammalian (red blood cells) evolved when O₂ levels were lower/ ORA (1) 2. {Biconcave/shape} increases surface area (for increased O₂ absorption) (1) 3. No nucleus + can carry more haemoglobin (so increased O₂ transport) / ORA(1) 4. Thin centres/small/ biconcave so short diffusion distance (so faster diffusion) (1) 5. Small(er) in size so have a higher sa : vol ratio (1) 6. Small(er), {so more of them/ total surface area larger} (1) Accept ref. to smaller capillaries qualified/ biconcave shape gives flexibility to fit through capillaries		4		4		
			Question 1 total	7	7	0	14	1	0